Sex-Specific Risk Factors and 10-Year Stroke Probability in the Framingham Heart Study

A Survival Analysis

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Background & Objectives

Background

- Framingham Heart Study (1948)
- First prospective cohort to identify CVD risk factors
- 5,209 community-dwelling adults; biennial examinations
- Dataset: 4,434 participants across 3 exam periods ($\sim$1956–1968)
- Outcomes: stroke, MI, angina, heart failure, death
- Analytic sample: $n = 4{,}275$ (complete case)

Study Objectives

Primary:

- Identify baseline risk factors associated with time to first stroke within 10 years, stratified by sex

Secondary:

- Estimate 10-year stroke-free survival under clinically meaningful risk profiles
- Summarize SBP and diabetes trends across exam periods to motivate time-varying covariate analyses

Methods

Data Management

- Period 1 only; excluded prior stroke ($n = 4{,}402$)
- Complete case analysis ($n = 4{,}275$)
- Outcome: time to stroke, capped at 3,652.5 days (10 yrs)
- Censored: stroke-free >10 yrs or non-stroke events
- Covariates: age, SBP, BMI, cholesterol, diabetes, smoking, BP meds, CHD

Variable Selection

- LASSO Cox regression, 10-fold CV (`glmnet`)
- Forced: Age, SBP, Diabetes (penalty = 0)
- Optional: BMI, cholesterol, CHD, BP meds, smoking
- λ_{min} used for variable selection
- Same covariate set applied to both sex-specific models

Model Fitting & Diagnostics

- Cox proportional hazards model (separately by sex)
- VIF assessed for multicollinearity
- PH assumption: Schoenfeld residuals
- 10-yr survival estimated at ages 40, 50, 60 under 5 risk profiles
- All analyses: R v4.3.1

Descriptive Results

Table 1: Baseline Characteristics by Sex

Characteristic	Male ($n = 1{,}930$)	Female ($n = 2{,}472$)
Age, mean (SD)	49.7 (8.71)	50.0 (8.63)
Systolic BP, mean (SD)	132 (19.4)	134 (24.3)
Coronary Heart Disease	6.2%	2.7%
Diabetes	3.1%	2.4%
Current Smoker	60.5%	40.5%
Stroke within 10 yrs	13.5%	9.3%

Table 2: SBP & Diabetes Trends Across Exam Periods

Period	Sex	SBP Mean (SD)	Diabetes %
Period 1	Male	131.6 (19.4)	3.1%
Period 1	Female	133.7 (24.3)	2.4%
Period 2	Male	135.4 (19.8)	4.4%
Period 2	Female	138.0 (24.3)	3.5%
Period 3	Male	139.3 (21.1)	8.7%
Period 3	Female	140.9 (24.1)	7.1%

Key Observations

KM curves confirmed:

- Diabetes $\rightarrow$ significantly lower stroke-free survival
- High SBP ($\geq$160 mmHg) $\rightarrow$ significantly lower survival

Temporal trends:

- SBP rose $\sim$8 mmHg over 3 periods in both sexes
- Diabetes prevalence nearly tripled in males (3.1% $\rightarrow$ 8.7%)
- Suggests baseline-only models may underestimate true risk

Cox Proportional Hazards Models

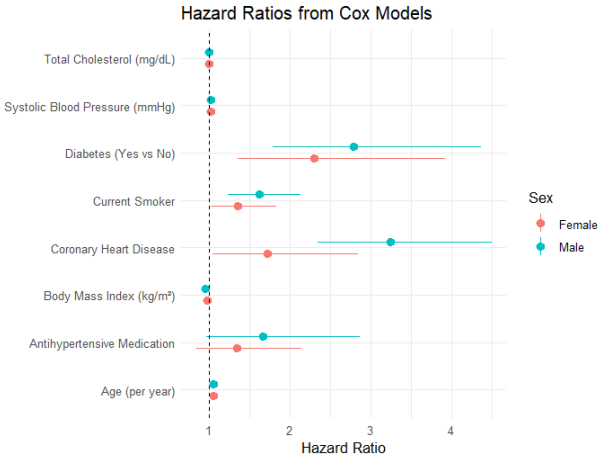

Figure: Hazard ratios from sex-specific Cox models with 95% CIs

Key Findings

Significant in both sexes: Age, SBP, Diabetes, CHD, Smoking

Males — stronger effects:

- CHD: HR 3.25 (95% CI: 2.35–4.50)
- Diabetes: HR 2.79 (95% CI: 1.79–4.37)
- BMI significant in males only

Females — smaller effects:

- CHD: HR 1.72 (95% CI: 1.04–2.85)
- Diabetes: HR 2.30 (95% CI: 1.35–3.92)

Diagnostics:

- All VIFs < 1.5 (no multicollinearity)
- PH assumption largely satisfied; mild deviation for smoking & SBP (females)

10-Year Stroke-Free Survival Probabilities

Males

Age	Baseline	High SBP	Diabetes	High SBP + Diabetes	Smoker
40	96.0%	92.4%	89.2%	80.1%	93.6%
50	93.3%	87.5%	82.4%	68.8%	89.4%
60	88.9%	79.7%	72.1%	**53.1%**	82.7%

Females

Age	Baseline	High SBP	Diabetes	High SBP + Diabetes	Smoker
40	97.0%	95.4%	93.2%	89.7%	95.9%
50	94.4%	91.6%	87.6%	81.7%	92.5%
60	89.8%	84.9%	78.1%	**68.5%**	86.4%

53.1%

Male, age 60, High SBP + Diabetes

68.5%

Female, age 60, High SBP + Diabetes

∼8 pp

Reduction from smoking (age 60, both sexes)

Conclusions

Key Findings

- Age, SBP, diabetes, CHD, and smoking are consistent stroke predictors in both sexes
- CHD and diabetes carry notably larger hazard ratios in males than females
- Combined high SBP + diabetes yields the greatest survival reduction, especially at older ages
- SBP and diabetes increased substantially across exam periods — baseline-only models may underestimate true risk
- Time-varying covariate analyses are warranted in future work

Limitations

- Complete case analysis assumes data are missing completely at random
- Competing risks not modeled — may overestimate stroke-free survival
- Demographically homogeneous cohort limits generalizability
- Baseline exposures assumed constant; mild PH violations noted for smoking and SBP in females